OpenRank Visualization Platform: System Architecture

LLM Open-Source Ecosystem Health Analytics Based on OpenDigger Metrics

1 User & Client

Web Browser

Student / Analyst / Researcher

Interactive Dashboard Access

HTTPS Request

Rendered Views

2 Vue 3 Frontend

Vite Build System

Vue Router

Pinia State Management

API Service Layer

ECharts Visualization Engine

Main Views

Home Dashboard

LLM Ecosystem Summary

Visual Lab

Favorites

Login / Register

Reusable Components

Metric Line Chart

Radar Compare

Contributor Bar Chart

Receives JSON Data from Backend

Renders Charts & Dashboard Views

REST API Requests (JSON)

JSON Responses (Data, Reports, AI Insights, Favorites)

3 Flask Backend Service

Application Entry: main.py

REST APIs

OpenDigger API Module	/api/llm/summary
LLM Summary / Report API	/api/llm/rank/metrics
Contributor Risk API	/api/llm/report
Auth API	/api/auth/login
Auth API (Register)	/api/auth/register
Favorites API	/api/favorites
Contributor Risk API	/api/health/contributors/risk

Middleware / Services

JWT Authentication

API Rate Limiting

CORS

Request Cache

Process Requests

Cached Data / DB Results

Query / Read (Results)

4 Data Processing & Storage

Data Processing Pipeline

Data Fetcher

Metric Utilities

Config Loader

Health Score Calculation

Time-Series Aggregation

Data Cache

Health Score =

0.5 x OpenRank + 0.3 x Activity + 0.2 x Stability

Processed Metrics (Parsed)

SQLite Database

User

Favorite

MetricSeries

Fetch Data (API / Scrape)

5 External Data Sources

OpenDigger

OpenRank Metrics

GitHub Repository Metadata

Project Activity Metrics

Developer / Contributor Metrics

Metric Ingestion

OpenRank | Activity

Stars | Forks

Issues | Pull Requests

Contributors

Platform Capabilities

- Interactive Visualization
- AI Interpretation
- PNG Export
- Report Export
- Favorites Management
- Risk Analysis
- Health Ranking
- Multi-chart Visual Analytics

Legend: Infrastructure / Backend · Data Processing · Analytics / AI Interpretation · Visualization / Capabilities

Primary Data Flow · Response / Return Flow · Bidirectional / Data Access